

1.1 INTRODUCTION

An organization is a structured group of individuals who work together to achieve common goals and objectives. Organizations can be business enterprises, educational institutions, government bodies, or non-profit entities. Every organization operates with the help of various resources such as financial resources, technological resources, and human resources. Among these, human resources are considered the most valuable asset because they directly contribute to productivity and organizational success. Organizations are established to fulfill specific purposes, and their success depends on effective coordination between employees, management, and operational systems. In today's competitive environment, organizations focus not only on profit but also on sustainability, innovation, and employee satisfaction.

1.2 IMPORTANCE OF EMPLOYEES IN AN ORGANIZATION

Employees are the backbone of any organization. They are responsible for executing tasks, achieving targets, and maintaining the quality of products and services. Without employees, no organization can function effectively. Employees bring skills, knowledge, creativity, and innovation, which are essential for organizational growth. The performance of employees directly influences customer satisfaction, organizational reputation, and financial success. Therefore, organizations invest heavily in recruitment, training, and development to ensure that employees perform efficiently. Retaining skilled employees has become a major challenge, making employee management a critical function of human resource management.

1.3 CONCEPT OF EMPLOYEE ATTRITION

Employee attrition is a natural process that occurs in every organization. It refers to the loss of employees due to various reasons such as retirement, resignation, or personal circumstances. Unlike employee turnover, attrition does not involve replacing the vacant position, which leads to a reduction in the workforce.

Attrition can be voluntary or involuntary. Voluntary attrition occurs when employees choose to leave, while involuntary attrition happens due to organizational decisions or unavoidable circumstances. Understanding the nature of attrition helps organizations manage workforce planning effectively.

1.4 IMPORTANCE OF STUDYING EMPLOYEE ATTRITION

Employee attrition is a critical issue because it affects organizational performance in multiple ways. When employees leave, it disrupts ongoing work, reduces productivity, and increases the workload of remaining employees. It also leads to additional costs related to recruitment, training, and development of new employees. Moreover, attrition results in the loss of valuable knowledge and experience, which can impact decision-making and innovation. Therefore, studying employee attrition helps organizations identify the root causes and develop strategies to reduce its negative impact.

1.5 CAUSES OF EMPLOYEE ATTRITION

Employee attrition occurs due to a variety of reasons. These reasons can be broadly categorized into employee-related factors, organizational factors, and external factors. Employee-related factors include personal issues, health problems, career aspirations, and work-life balance. Organizational factors include low salary, poor working conditions, lack of recognition, and ineffective management. External factors such as better job opportunities, economic conditions, and industry trends also influence employee decisions. Understanding these causes is essential for organizations to design effective retention strategies and improve employee satisfaction.

1.6 POSITIVE AND NEGATIVE IMPACT OF ATTRITION

Employee attrition has both positive and negative impacts on an organization. On the negative side, it leads to loss of skilled employees, increased costs, and reduced productivity. It may also affect team morale and organizational culture.

However, attrition can also have positive effects. It can reduce labor costs, create opportunities for restructuring, and bring new ideas and perspectives into the organization. Controlled attrition can help organizations adapt to changing business environments and improve efficiency.

1.7 EMPLOYEE ATTRITION PROCESS

The process of employee attrition typically begins with employee dissatisfaction, followed by the decision to leave the organization. Once the decision is made, the employee submits a resignation, and the organization initiates the exit process. This

includes exit interviews, documentation, and knowledge transfer. Managing this process effectively is important to minimize disruption and ensure a smooth transition. Organizations can use exit feedback to identify issues and improve their policies and practices.

1.8 FACTORS USED IN ATTRITION PREDICTION

Employee attrition prediction involves analyzing various factors such as age, gender, salary, job role, work experience, job satisfaction, and work environment. These factors provide insights into employee behavior and help in identifying patterns that lead to attrition.

By analyzing these factors, organizations can develop strategies to improve employee engagement and reduce the likelihood of attrition.

1.9 NEED FOR EMPLOYEE ATTRITION PREDICTION

Traditionally, organizations addressed attrition after it occurred. However, this reactive approach is not effective in today's dynamic environment. With advancements in technology, it is now possible to predict employee attrition in advance using data analysis techniques.

Attrition prediction helps organizations identify employees who are at risk of leaving and take proactive measures to retain them. This reduces employee loss and improves overall organizational performance.

1.10 BENEFITS OF ATTRITION PREDICTION

Attrition prediction offers several benefits to organizations. It helps in reducing employee turnover, saving recruitment and training costs, and improving employee satisfaction. It also enhances productivity and supports better decision-making. By identifying potential attrition risks in advance, organizations can implement effective retention strategies and maintain a stable workforce.

1.11 EMPLOYEE ATTRITION AND RETENTION

In human resource management, employee attrition and retention are two important concepts. Employee attrition refers to the gradual reduction in the number of employees when they leave the organization and are not replaced. This can occur due to retirement, personal reasons, or career changes. Employee retention, on the other hand, refers to

the ability of an organization to retain its employees for a longer period by maintaining job satisfaction and engagement.

Both attrition and retention are interconnected. High attrition indicates poor retention strategies, while effective retention reduces unnecessary employee loss. Organizations aim to maintain a balance between these two to ensure stability and growth.

1.12 ATTRITION REASONS ACROSS DIFFERENT SECTORS

Attrition reasons differ across sectors based on working conditions and employee expectations. Common reasons include low salary, work pressure, lack of career growth, poor work-life balance, and job dissatisfaction. In IT and service sectors, stress and better job opportunities increase attrition. In manufacturing and retail sectors, low wages and limited promotions are major causes. Overall, employee dissatisfaction and better external opportunities are the key reasons for attrition.

1.12.1. IT SECTOR (INFORMATION TECHNOLOGY)

The Information Technology (IT) sector is one of the most dynamic and fast-growing industries, but it also faces a high rate of employee attrition. One of the major reasons for attrition in this sector is the availability of better job opportunities. Skilled IT professionals are always in demand, and employees frequently switch jobs for higher salaries, better roles, and career growth. Another important reason is work pressure and long working hours. Many IT employees experience stress due to tight deadlines, project demands, and continuous performance expectations.

Additionally, lack of work-life balance plays a significant role in employee attrition. Employees often struggle to manage personal and professional life, especially in roles that require night shifts or extended working hours. Limited career growth or lack of learning opportunities also leads employees to leave organizations. In some cases, poor management, lack of recognition, and ineffective leadership further contribute to dissatisfaction.

The rapid technological changes in the IT industry also force employees to constantly upgrade their skills. If organizations fail to provide proper training and development opportunities, employees may leave in search of better learning environments. Thus,

high competition, stress, and growth expectations are the primary drivers of attrition in the IT sector.

1.12.2 HEALTHCARE SECTOR

The healthcare sector faces unique challenges when it comes to employee attrition. One of the primary reasons is high work stress and emotional pressure. Healthcare professionals such as doctors, nurses, and support staff deal with critical situations, long working hours, and patient care responsibilities, which can lead to burnout.

Another major factor is workload imbalance. Many healthcare institutions face staff shortages, which increases the burden on existing employees. This often results in fatigue, dissatisfaction, and eventually resignation. In addition, safety concerns and exposure to health risks, especially after events like pandemics, have increased attrition rates in this sector.

Low compensation compared to workload is also a significant issue, particularly for nurses and support staff. Many employees feel that their efforts are not adequately rewarded. Lack of career advancement opportunities and limited professional development can further demotivate employees.

Moreover, work-life balance is difficult to maintain in healthcare due to irregular shifts and emergency duties. Emotional exhaustion and lack of organizational support also contribute to employees leaving the profession or shifting to less demanding roles. Therefore, stress, workload, and safety concerns are the key reasons for attrition in the healthcare sector.

1.12.3. MANUFACTURING SECTOR

The manufacturing sector experiences attrition due to a combination of physical, economic, and organizational factors. One of the main reasons is the physically demanding nature of the work. Employees often work in challenging environments with long hours, repetitive tasks, and strict production targets, which can lead to fatigue and job dissatisfaction.

Low wages and limited financial incentives are also major contributors to attrition in this sector. Many workers leave in search of better-paying jobs or more stable employment opportunities. Additionally, lack of job security, especially in contract-based or temporary roles, increases employee turnover.

Another important factor is limited career growth. Employees in manufacturing often have fewer opportunities for promotion or skill development compared to other sectors. This leads to a lack of motivation and encourages employees to switch jobs.

Workplace conditions and safety concerns also play a crucial role. Poor working environments, lack of proper safety measures, and risk of accidents can push employees to leave. Furthermore, migration of workers to urban areas or other industries for better opportunities contributes to attrition.

In recent times, automation and technological advancements have also impacted employment stability, causing uncertainty among workers. As a result, physical strain, low pay, and lack of growth opportunities are the primary reasons for attrition in the manufacturing sector.

REVIEW OF LITERATURE

Rahul Yedida et al., 2018 The study focused on predicting employee attrition using the k-Nearest Neighbors (KNN) algorithm. Key features such as employee performance, average monthly working hours, and tenure in the organization were used for analysis. The model was trained and tested using a 70:30 data split and achieved a high accuracy of 94.32%. The research also highlighted alternative approaches like artificial neural networks, decision trees, and logistic regression for comparison.

Francesca Fallucchi et al., 2020 This study examined the role of artificial intelligence in supporting decision-making within human resource management, particularly in predicting employee attrition. Using a real-world dataset from IBM analytics with 35 features and around 1500 samples, the research focused on identifying objective factors influencing employee turnover. Among various algorithms tested, the Gaussian Naïve Bayes classifier achieved the best performance, showing a higher recall rate and effectively minimizing false negatives. The findings emphasize the importance of data-driven approaches in improving HR decisions and predicting employee behaviour.

Ali Raza et al., 2022 The study focused on analysing the key organizational factors contributing to employee attrition and predicting it using machine learning techniques. Multiple models were compared, and the optimized Extra Trees Classifier (ETC) achieved the highest accuracy of 93%. The research also utilized exploratory data analysis to identify important factors such as monthly income, hourly rate, job level, and age influencing attrition. The findings highlight how data-driven approaches can help organizations reduce employee turnover and improve retention strategies.

R Shiva Shankar et al., 2018 This study addressed employee attrition as a critical issue affecting organizational performance and costs. The researchers applied multiple machine learning classification techniques, including Decision Tree, Logistic Regression, Support Vector Machine (SVM), KNN, Random Forest, and Naïve Bayes, along with feature selection methods to improve prediction accuracy. By analysing both current and past employee data, the study identified common reasons for attrition and emphasized the importance of predictive models in reducing employee turnover and associated costs.

Haya Alqahtani et al., 2024 This study presented a comprehensive literature review on the application of machine learning techniques for employee attrition prediction from 2019 to 2024. The research analysed and classified various studies, highlighting that ensemble methods such as Random Forest and XGBoost consistently deliver superior performance compared to other models. It also noted the growing potential of deep learning approaches and the widespread use of the IBM dataset in attrition research. The study provides valuable insights into key factors influencing attrition and guides researchers and organizations in selecting effective predictive models.

M. Subhashini and R. Gopinath, 2020 This study focused on improving employee attrition prediction using advanced feature selection and machine learning techniques. Various feature selection methods, including Chi-Square, Information Gain, and Gain Ratio, were evaluated alongside classifiers such as Artificial Neural Networks, Support Vector Machines, Gradient Boosting, Random Forest, and Decision Trees. The findings revealed that the combination of Chi-Square feature selection with Gradient Boosting Tree achieved higher accuracy and reduced error rates. The study emphasizes the importance of selecting relevant features to enhance prediction performance.

Salah Al-Darraj et al., 2021 This study explored the use of deep learning techniques to improve employee attrition prediction. Using the IBM analytics dataset, the researchers analysed various factors influencing attrition and addressed data imbalance by creating a balanced dataset. The model was evaluated using cross-validation, achieving an accuracy of 91% on the original dataset and 94% on the balanced dataset. The findings highlight the effectiveness of deep learning and data pre-processing techniques in enhancing prediction accuracy and identifying key attrition factors.

Sarah S. Alduayj and Kashif Rajpoot, 2018 This study investigated employee attrition prediction using multiple machine learning models on an IBM Watson synthetic dataset. The research addressed class imbalance using techniques such as ADASYN and under sampling, which significantly improved model performance. Among the models tested, K-Nearest Neighbours (KNN) achieved the highest performance with an F1-score of 0.93 when trained on the balanced dataset. Additionally, feature selection combined with Random Forest also showed strong predictive capability, highlighting the importance of data balancing and feature optimization.

Samer M. Arqawi et al., 2022 This study focused on predicting employee attrition using both machine learning and deep learning techniques. The researchers used an IBM-based dataset from Kaggle and applied pre-processing and data balancing methods to improve model performance. The deep learning model outperformed traditional machine learning models, achieving higher accuracy, precision, recall, and F1-score. The study also emphasized identifying key factors influencing attrition, demonstrating the effectiveness of advanced learning models in improving prediction accuracy.

Nesrine Ben Yahia et al., 2021 This study proposed a people analytics approach to predict employee attrition by emphasizing data quality over quantity through a “deep data” methodology. The research combined exploratory and quantitative methods, including feature selection and survey analysis, to identify the most relevant factors influencing attrition. Multiple models, including machine learning, deep learning, and ensemble techniques, were applied across different datasets, achieving high accuracy levels up to 99%. The study also revealed that factors like business travel play a significant role in employee retention, offering new insights for HR policies.

Sandeep Yadav, Aman Jain, and Deepti Singh (2018) conducted a study on employee attrition and its impact on organizational performance. The study highlighted that employee turnover creates significant costs for organizations and affects overall efficiency and productivity. The authors emphasized that replacing skilled and experienced employees is a difficult and expensive process for companies. The research focused on predicting employee churn by analysing employee behaviour and attributes using classification techniques. The study also suggested that proper attrition prediction can help organizations reduce turnover and improve employee retention strategies.

Namrata Bhartiya, Sheetal Jannu, Purvika Shukla, and Radhika Chapaneri (2019) conducted a study on employee attrition prediction using Data Analysis and Machine Learning techniques. The study explained a systematic process involving data acquisition, data conditioning, visualization, and classification methods to identify attrition patterns. Various classification models such as Support Vector Machine, Decision Tree, K-Nearest Neighbour, Random Forest, and Naive Bayes were applied in the Python environment. The results showed that the Random Forest classifier achieved the highest accuracy of 83.3%, while Naive Bayes and Support Vector

Machine performed better in identifying true positive cases. The study concluded that machine learning techniques can effectively help organizations identify the major causes of attrition and support employee retention strategies.

Nesreen El-Rayes, Ming Fang, Michael Smith, and Stephen M. Taylor (2020) conducted a study to predict employee attrition using tree-based binary classification models. The research focused on analyzing firm culture and management-related factors that influence employee turnover. The study used anonymous resume data collected from Glass door along with public company review information to develop prediction models. Decision Tree, Random Forest, and Gradient Boosted Tree models were applied to estimate the probability of employees leaving an organization during job transitions. The study concluded that organizational culture and management practices play an important role in predicting and reducing employee attrition.

Fahad Kamal Alsheref, Ibrahim Eldesouky Fattoh, and Waleed M. Ead (2022) conducted a study on predicting employee attrition using predictive analytics and machine learning techniques. The study highlighted that retaining skilled and experienced employees is a major challenge for organizations, as employee turnover leads to increased costs and loss of expertise. The researchers developed an automated prediction model using different analytical techniques and pipeline architectures to identify the most effective model. An auto tuning approach was also applied to determine the best hyper parameter combinations for improving model performance. The study concluded that although no single model is perfect for every business situation, the proposed ensemble model effectively supported employee attrition prediction and organizational decision-making.

Manju Nandal, Veena Grover, Divya Sahu, and Mahima Dogra (2024) studied employee attrition prediction using machine learning and deep learning techniques. Various models such as Logistic Regression, Random Forest, XG Boost, FNN, and CNN were applied to the IBM Watson dataset. The study found that the Feed forward Neural Network (FNN) achieved the highest prediction accuracy of 97.5%. The research concluded that deep learning models can effectively help organizations predict and reduce employee attrition.

CHAPTER 3

3.1 DESIGN OF THE STUDY

The study on **Prediction of Employee Attrition** follows a **descriptive and analytical research design** with a quantitative approach. Primary data was collected from **150 employees** using a structured questionnaire distributed via **Google Forms**, capturing variables such as age, salary, job satisfaction, work-life balance, and intent to leave. Secondary data was sourced from research journals, HR reports, and publicly available datasets to support the theoretical background. A **purposive sampling method** was adopted to ensure respondents had relevant work experience. The collected data was pre-processed and analysed using three machine learning algorithms — **Decision Tree, Random Forest, and Artificial Neural Network (ANN)** — implemented in Python. Model performance was evaluated using accuracy, precision, recall, F1-score, and AUC-ROC to identify the most effective algorithm for predicting employee attrition.

3.1.1 RESEARCH DESIGN

The present study adopts a descriptive and analytical research design. A descriptive design is employed to describe the characteristics and patterns of employee attrition in the selected sample population. The analytical component involves applying machine learning techniques — Decision Tree, Random Forest, and Artificial Neural Network — to analyze the data and build predictive models.

This design is appropriate as the study seeks to both describe the demographic and work-related attributes of employees and to predict future attrition outcomes based on those attributes. The research follows a structured, step-by-step methodology starting from data collection, followed by data pre-processing, model building, model evaluation, and result interpretation.

The research is quantitative in nature, relying on numerical data and statistical analysis to draw conclusions. The study uses a cross-sectional design, meaning that data is collected at a single point in time from a selected sample of employees across various departments and roles within an organization.

3.1.2 NATURE OF DATA

- **Primary data**

The primary data is defined as the data, which is collected for the first time and fresh in nature, and happen to be original in character through Google forms.

- **Secondary data**

The secondary data are those which have already been collected by someone else and have been passed through statistical process. The secondary data has been collected from articles, journals, thesis and web-based resource.

3.1.3 METHODOLOGY

The data collected through Google Forms was exported as a **CSV file** and imported into **IBM SPSS Modeler** for analysis. The **Type node** was used to define input and target variables, and the **Partition node** was used to split the data into **80% training and 20% testing**. Three machine learning models were built using the **C5.0 node** (Decision Tree), **Random Forest node**, and **Neural Network node** (ANN). The models were evaluated using the **Analysis node** and **Evaluation node**, generating metrics such as **Accuracy, Precision, Recall, F1-Score, and AUC-ROC** to compare and identify the best performing model for predicting employee attrition.

3.1.3.1 SAMPLE SIZE

The primary data has been collected through Google forms method. Google forms Survey has been Conducted using well formulated questionnaire. Samples for the purpose of this study have been selected systematically. The questionnaire was sent to 250 peoples who overuse digital presence; the researcher received 188 responses only. All the 188 responses were considered for the study as samples.

3.1.3.2 SAMPLING METHOD

The study employed a non-probability convenience sampling method to select respondents. Convenience sampling was chosen because the study required participants who are currently employed or have recently left an organization and can provide relevant information about workplace factors that influence attrition. This method

ensures that the sample consists of individuals who are most informative for the research objectives.

Additionally, convenience sampling was adopted for practical reasons, as respondents were selected from among those who were accessible and willing to participate. The researcher ensured diversity in the sample by targeting employees across different age groups, departments, educational qualifications, income brackets, and levels of work experience to make the findings as representative as possible.

3.1.3.3 METHOD OF DATA COLLECTION

The primary data for this study was collected using Google Forms, an online survey tool. A structured questionnaire was created and distributed electronically to the target respondents via email, WhatsApp, and professional networking platforms such as LinkedIn. Google Forms was chosen for its ease of use, wide reach, low cost, and ability to automatically organize responses into a structured format that is compatible with data analysis tools.

The questionnaire was divided into five sections: (i) demographic information, (ii) job and role-related attributes, (iii) compensation and benefits, (iv) workplace environment and satisfaction, and (v) intentions to leave. A five-point Likert scale was used for attitudinal variables, while categorical and numerical scales were used for factual questions.

Before distributing the questionnaire, a pilot study was conducted with a small group of 15 respondents to test the clarity, relevance, and reliability of the questions. Feedback from the pilot study was used to refine and improve the questionnaire. The final data collection phase was carried out over a period of three weeks.

3.1.3.4 DATA ANALYSIS AND RESEARCH

The methodology adopted for the study on **Prediction of Employee Attrition** involves a structured, data-driven approach using machine learning techniques. A sample of **188 employees** was selected through purposive sampling, and data was collected via a structured questionnaire distributed through **Google Forms**. The questionnaire covered key variables including demographics, job role, compensation, job satisfaction, and intention to leave. The collected data was pre-processed by handling missing values, encoding categorical variables, and normalising numerical features to ensure data

quality. Three machine learning models — **Decision Tree, Random Forest, and Artificial Neural Network (ANN)** — were built and analysed using **IBM SPSS Modeler**, a powerful predictive analytics tool that provides a visual interface for building, evaluating, and deploying machine learning models. The dataset was split in an **80:20 ratio**, with 80% used for training and 20% for testing. The models were evaluated and compared using performance metrics such as **Accuracy, Precision, Recall, F1-Score, and AUC-ROC** to determine the most reliable and accurate model for predicting employee attrition.

3.2.4 HYPOTHESIS FRAMED

The following hypotheses were formulated to guide the research and statistical analysis:

- H0₁: There is no significant relationship between employee age and employee attrition.
- H1₁: There is a significant relationship between employee age and employee attrition.
- H0₂: There is no significant relationship between monthly income and employee attrition.
- H1₂: There is a significant relationship between monthly income and employee attrition.
- H0₃: There is no significant relationship between job satisfaction and employee attrition.
- H1₃: There is a significant relationship between job satisfaction and employee attrition.
- H0₄: Decision Tree, Random Forest, and Artificial Neural Network do not differ significantly in predicting employee attrition.
- H1₄: There is a significant difference in the predictive accuracy of Decision Tree, Random Forest, and Artificial Neural Network in predicting employee attrition.
- H0₅: Work-life balance has no significant impact on the decision of an employee to leave the organization.

- H1₅: Work-life balance has a significant impact on the decision of an employee to leave the organization.

3.2.5 NEED FOR THE STUDY

Employee attrition is one of the most pressing challenges faced by organizations today. The cost of replacing an employee can range from 50% to 200% of their annual salary when considering recruitment, on boarding, training, and the productivity loss during the transition period. Despite this, many organizations continue to manage attrition reactively rather than proactively, addressing the problem only after employees have already resigned.

There is a growing need for data-driven approaches that can identify employees who are at risk of leaving before they actually do so. Machine learning offers a powerful and scalable solution to this challenge by analyzing patterns in historical employee data and generating predictive insights that HR departments can act upon in real time.

This study is needed because: (i) traditional HR management methods often rely on subjective judgment rather than objective data analysis; (ii) machine learning models can process large volumes of complex data and uncover non-obvious attrition risk factors; (iii) there is a need to compare and evaluate the performance of multiple machine learning algorithms specifically in the context of employee attrition prediction; (iv) organizations, especially in India, are increasingly investing in HR analytics but lack validated methodologies tailored to their workforce characteristics.

Furthermore, this study contributes to the growing body of academic literature on HR analytics and machine learning by providing empirical evidence on the effectiveness of Decision Tree, Random Forest, and Artificial Neural Network models in predicting employee attrition. The findings are expected to be of practical value to HR professionals, data scientists, and organizational leaders seeking to improve workforce stability.

3.2.6 LIMITATION OF THE STUDY

- The study is limited to a **sample size of 150 respondents**, which may not fully represent the entire working population across all industries.
- The data was collected through a **self-administered questionnaire**, which may be subject to response bias as respondents may not answer with complete honesty.
- The study is **cross-sectional in nature**, meaning data was collected at a single point in time and does not capture changes in employee behaviour over time.
- The research is confined to a **specific geographical region** and may not reflect the attrition patterns of employees in other regions or countries.
- The findings are based on **selected variables** and may not account for all possible factors influencing employee attrition, such as external market conditions or personal circumstances.
- The predictive models built using **IBM SPSS Modeler** are dependent on the quality and quantity of the data provided, and results may vary with larger or more diverse datasets.

3.2.7 RESEARCH OBJECTIVE

The objectives of this research study are as follows:

- To identify the key demographic and work-related factors that significantly influence employee attrition.
- To collect primary data from employees using a structured questionnaire and Google Forms to understand the underlying causes of attrition.
- To build predictive models for employee attrition using three machine learning algorithms: Decision Tree, Random Forest, and Artificial Neural Network (ANN).
- To compare the performance of the Decision Tree, Random Forest, and ANN models in terms of accuracy, precision, recall, F1-score, and AUC-ROC.
- To determine the most accurate and reliable machine learning model for predicting employee attrition in an organizational setting.

- To provide actionable recommendations to HR professionals and management to reduce employee attrition based on the insights generated by the predictive models.
- To contribute to the body of knowledge in the field of HR analytics and machine learning by providing a comparative empirical study on employee attrition prediction.

3.2.8 CHAPTERISATION

Chapter I - Introduction

This chapter includes meaning of Perceived Stress. Determinants and Outcomes of Self efficacy, Impact on Turnover Intention.

Chapter II - Review of Literature

This chapter provides only Review of Literature of this related study.

Chapter 3 – Research Methodology

This chapter outlines the research design, data collection methods, sampling procedure, and analytical framework adopted in the study. It details the use of primary data collected via Google Forms, the application of machine learning algorithms, and the statistical tools used to evaluate model performance. The chapter also presents the hypotheses, research objectives, limitations, and the need for the study.

Chapter 1V - Analysis and interpretation

This chapter covers data interpretation & test for the study. It includes ANOVA, Correlation and regression.

Chapter V - Summary of Findings, Suggestions and Conclusion

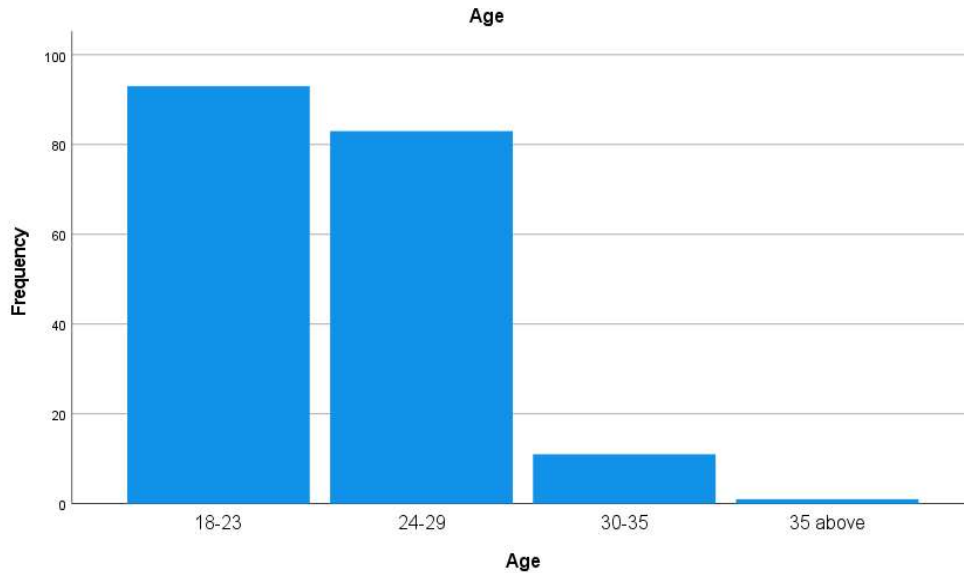
The final chapter is findings of the study and provides suggestions, conclusion on determinants and outcomes of switching factor impact on life satisfaction.

4.1 FREQUENCY TABLE

4.1.1 Result of frequency analyses test for age of respondents

Age					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	18-23	93	49.5	49.5	49.5
	24-29	83	44.1	44.1	93.6
	30-35	11	5.9	5.9	99.5
	35 above	1	.5	.5	100.0
	Total	188	100.0	100.0	

4.1.1. Frequency Table for Age



4.1.1. Chart for Age of Respondents

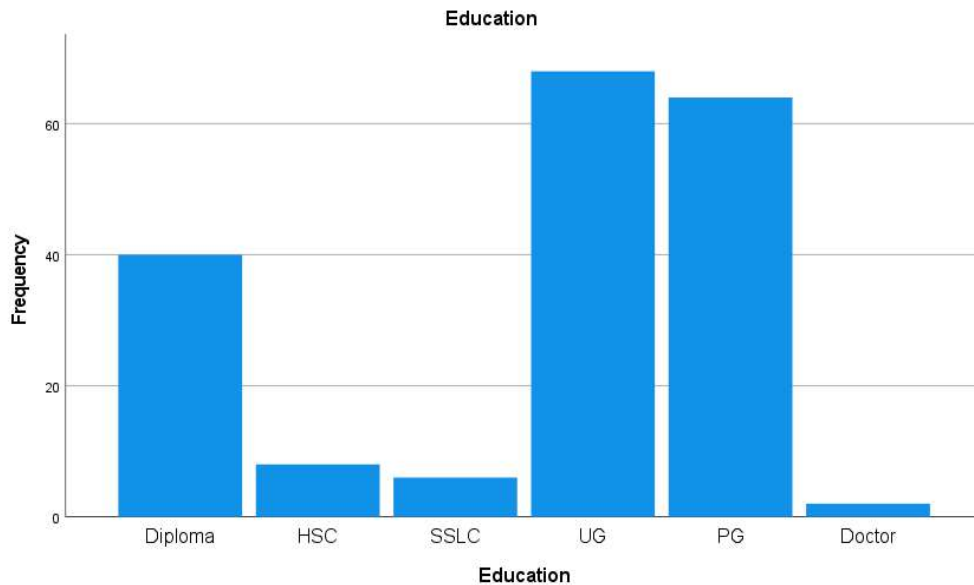
INTERPRETATION

The age distribution of the respondents shows that the majority belong to the younger age groups. Out of 188 respondents, 49.5% (93 respondents) are in the 18–23 age category, making it the largest group. This is followed by 44.1% (83 respondents) in the 24–29 age group. Only 5.9% (11 respondents) fall within the 30–35 age category, while respondents aged above 35 represent just 0.5% (1 respondent). Overall, the data indicates that the study mainly consists of young adults, with nearly 94% of respondents being below 30 years of age.

4.1.2. Result of frequency analyses test for Education of respondents

Education					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Diploma	40	21.3	21.3	21.3
	HSC	8	4.3	4.3	25.5
	SSLC	6	3.2	3.2	28.7
	UG	68	36.2	36.2	64.9
	PG	64	34.0	34.0	98.9
	Doctor	2	1.1	1.1	100.0
	Total	188	100.0	100.0	

4.1.2. Frequency Table for Gender



4.1.2. Chart for Education of respondents

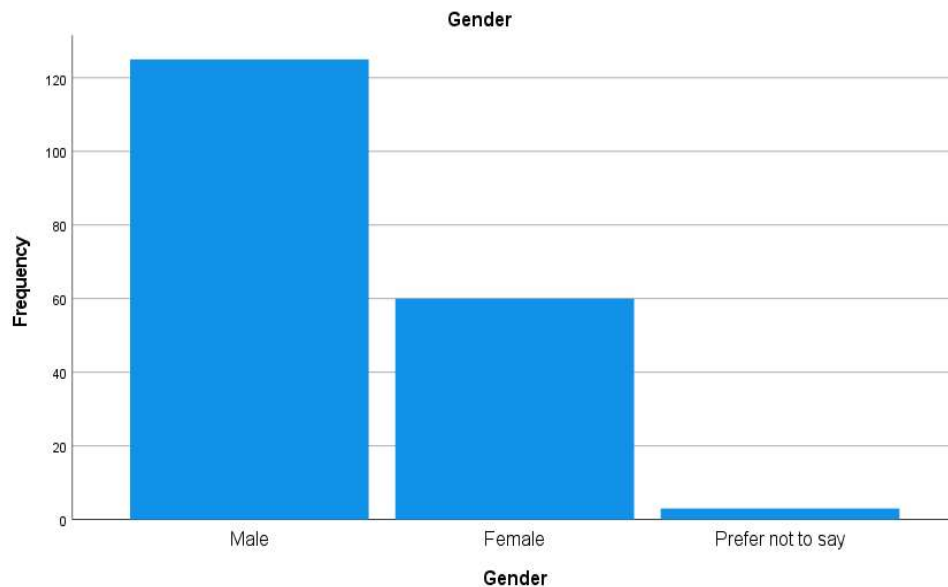
INTERPRETATION

The educational profile of the respondents indicates that most participants are highly educated. The majority have completed UG (36.2%) and PG (34.0%) qualifications, followed by Diploma holders (21.3%). Only a small percentage have SSLC, HSC, or Doctoral qualifications. Overall, the sample mainly consists of respondents with higher education backgrounds.

4.1.3. Result of frequency analyses test for gender of respondents

Gender					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Male	125	66.5	66.5	66.5
	Female	60	31.9	31.9	98.4
	Prefer not to say	3	1.6	1.6	100.0
	Total	188	100.0	100.0	

4.1.3. Frequency Table for Gender



4.1.3. Chart for Gender of Respondents

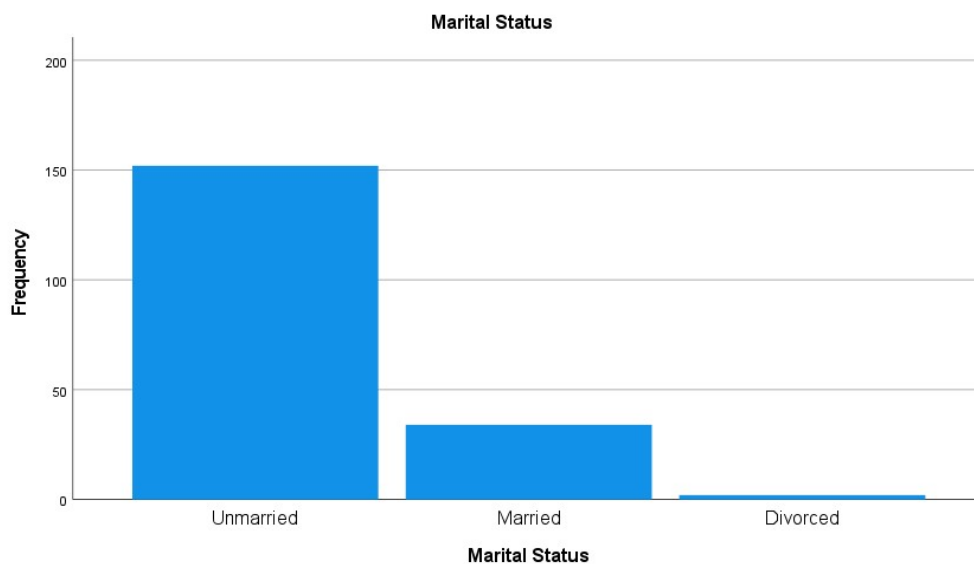
INTERPRETATION

The gender distribution of the sample (N=188) shows a clear majority of male participants, with 125 respondents (66.5%) identifying as male. Female respondents numbered 60, accounting for nearly one-third of the sample at 31.9%. A minimal proportion of participants—only 3 individuals, representing 1.6%—chose the option "Prefer not to say." Cumulatively, male and female respondents together make up 98.4% of the total sample, leaving a very small fraction who opted not to disclose. Overall, the data indicates a significantly imbalanced gender composition, with males outnumbering females by more than two to one. This skew should be considered when generalizing findings to a broader, more gender-balanced population.

4.1.4. Result of frequency analyses test for highest qualification of respondents

Marital Status					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Unmarried	152	80.9	80.9	80.9
	Married	34	18.1	18.1	98.9
	Divorced	2	1.1	1.1	100.0
	Total	188	100.0	100.0	

4.1.4. Frequency Table for Gender of Respondents



4.1.4. Chart for Gender of Respondents

INTERPRETATION

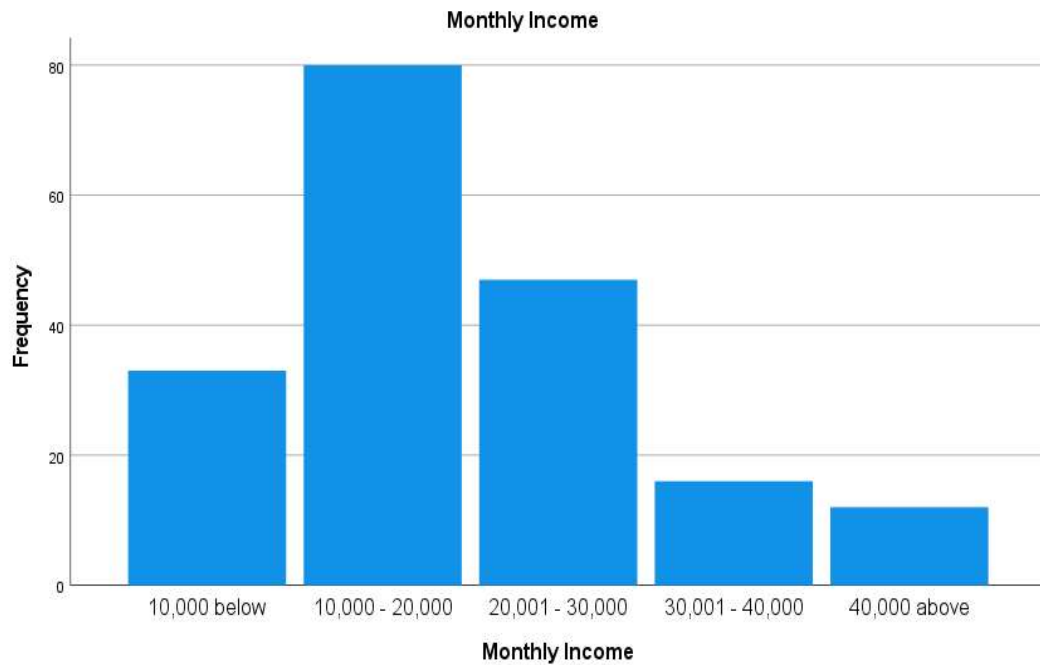
The marital status distribution among the 188 respondents reveals that the vast majority are unmarried. Specifically, 152 participants (80.9%) reported being unmarried, making this the dominant category in the sample. In contrast, married respondents account for only 34 individuals, representing 18.1% of the total. A very small fraction—just 2 respondents (1.1%)—reported being divorced. Cumulatively, unmarried and married participants together comprise 98.9% of the sample, leaving only a negligible percentage for the divorced category. Overall, the sample is heavily skewed toward unmarried individuals, with a ratio of more than 4 unmarried persons for every married person. This suggests the sample likely consists predominantly of younger adults or a

population where marriage is not yet common, and divorced individuals are virtually absent from the analysis.

4.1.5. Result of frequency analyses test for Monthly Income of respondents

Monthly Income					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	10,000 below	33	17.6	17.6	17.6
	10,000 - 20,000	80	42.6	42.6	60.1
	20,001 - 30,000	47	25.0	25.0	85.1
	30,001 - 40,000	16	8.5	8.5	93.6
	40,000 above	12	6.4	6.4	100.0
	Total	188	100.0	100.0	

4.1.5. Table for Monthly Income of Respondents



4.1.5. Chart for Monthly Income of Respondents

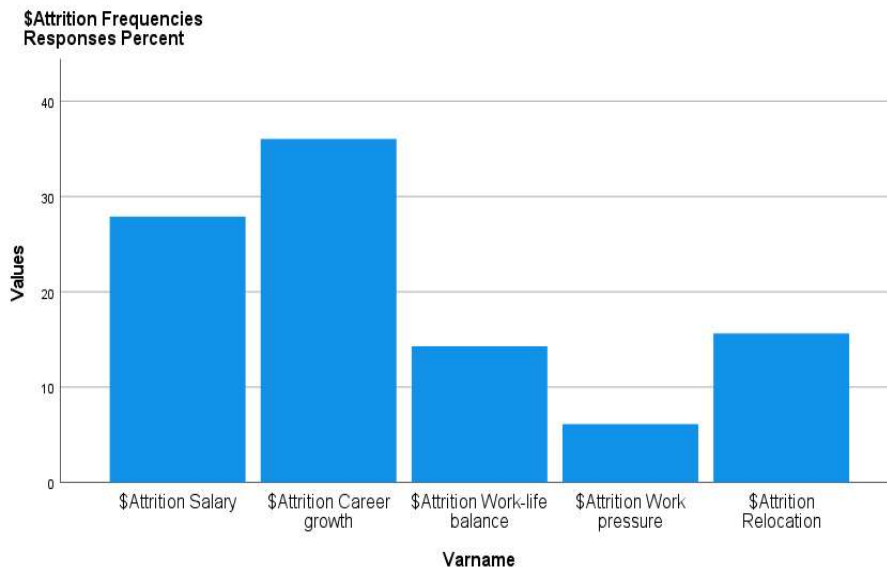
INTERPRETATION

The monthly income of respondents (N=188) is concentrated in the lower ranges. The largest segment (42.6%) earns between ₹10,000–₹20,000, followed by 25.0% earning ₹20,001–₹30,000. Those earning below ₹10,000 make up 17.6%. Only 15% of respondents earn above ₹30,000 (8.5% between ₹30,001–₹40,000 and 6.4% above ₹40,000). Cumulatively, over 85% earn ₹30,000 or less, indicating a predominantly low-to-middle income sample with very few high-income earners.

4.1.6. Result of frequency analyses test for Attrition Frequencies of respondents

\$Attrition Frequencies				
		Responses		Percent of Cases
		N	Percent	
\$Attrition ^a	Salary	41	27.9%	28.1%
	Career growth	53	36.1%	36.3%
	Work-life balance	21	14.3%	14.4%
	Work pressure	9	6.1%	6.2%
	Relocation	23	15.6%	15.8%
Total		147	100.0%	100.7%
a. Group				

4.1.6. Table for Monthly Income of Respondents



4.1.6. Chart for Monthly Income of Respondents

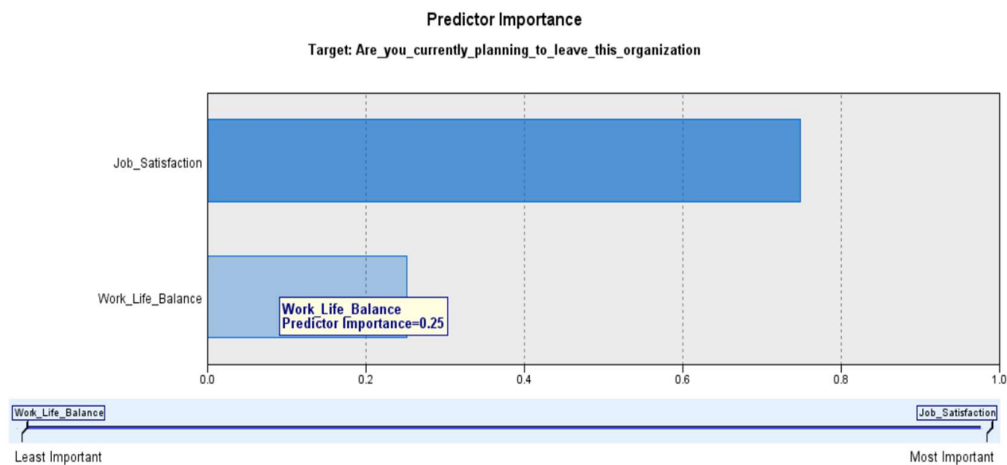
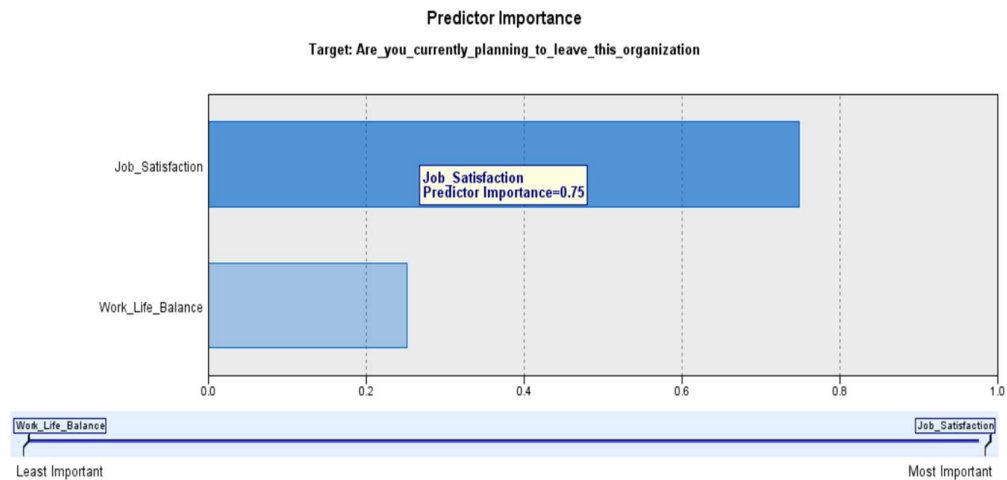
INTERPRETATION

Among the 188 respondents (with 147 total responses due to multiple selections allowed), the primary reason for attrition is **Career growth**, cited by 53 respondents (36.1% of responses, 36.3% of cases). The second most common reason is **Salary**, mentioned by 41 respondents (27.9% of responses, 28.1% of cases). **Relocation** and **Work-life balance** follow, with 23 (15.6%) and 21 (14.3%) responses respectively. **Work pressure** is the least cited factor, with only 9 responses (6.1%). Overall, career advancement and salary concerns together account for nearly two-thirds (64%) of all attrition reasons, suggesting that professional growth and compensation are the key drivers of employee turnover in this sample. Work-life balance and relocation play moderate roles, while work pressure is a relatively minor factor.

4.2 CHAID Decision Tree Analysis

The Chi-squared Automatic Interaction Detection (CHAID) algorithm was applied to identify the most significant predictors of employee attrition. The analysis reveals a hierarchical structure of factors that drive employees' intention to leave the organization.

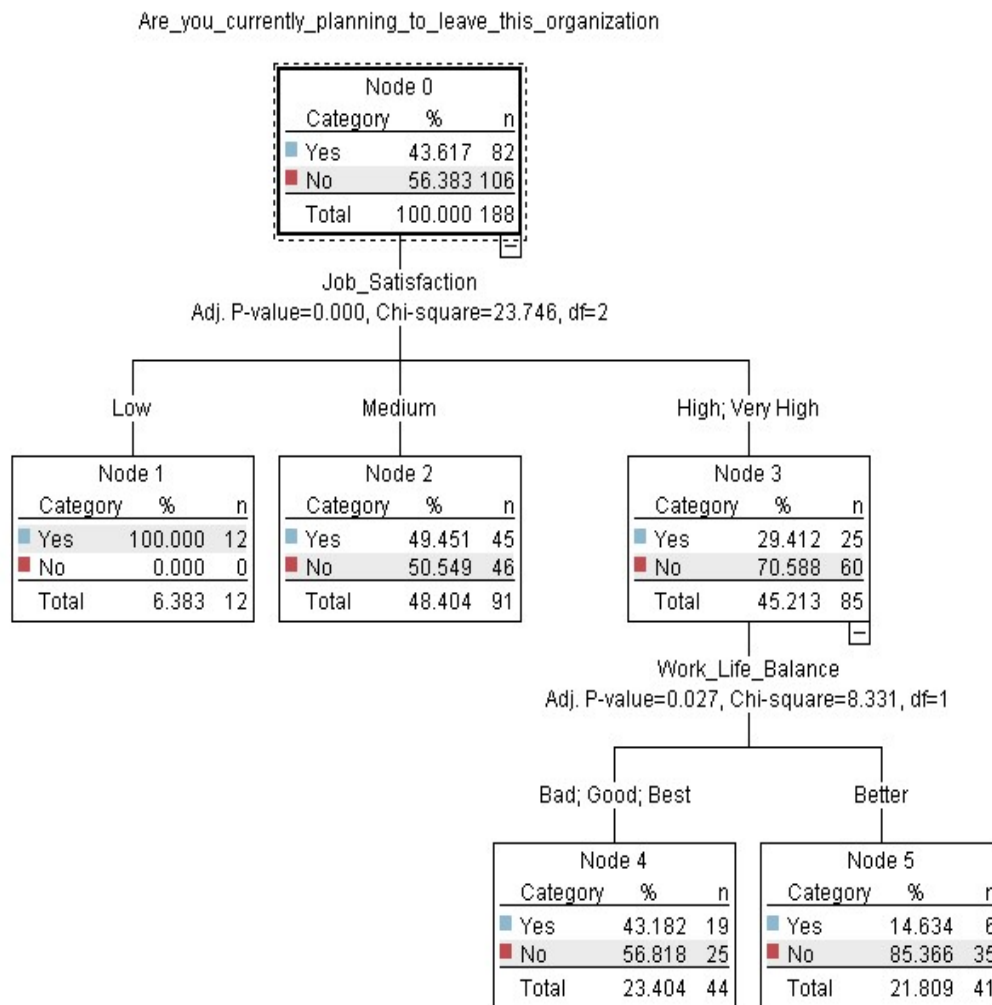
4.2.1 Predictor Importance — CHAID



INTERPRETATION

The predictor importance chart indicates that Job Satisfaction is the most influential factor affecting employees' intention to leave the organization with a predictor importance value of 0.75. Work-Life Balance has a lower influence (0.25), indicating a moderate secondary role in attrition decisions.

4.2.2 CHAID Decision Tree — Attrition Segments

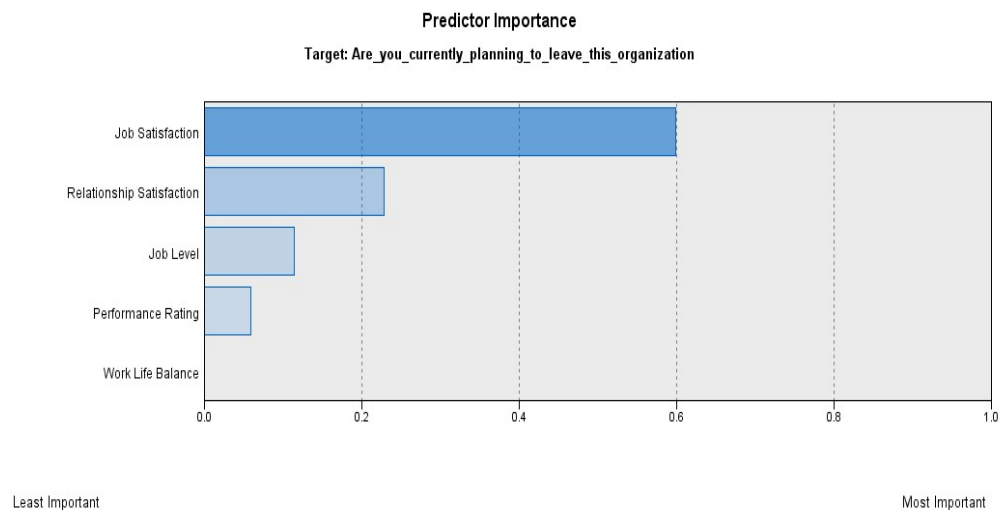


INTERPRETATION

The decision tree analysis shows that **Job Satisfaction** is the primary factor influencing employees' intention to leave the organization. Employees with low job satisfaction show a higher intention to leave, while those with high satisfaction are more likely to stay. Among highly satisfied employees, **Work-Life Balance** further affects retention, where better work-life balance reduces employees' intention to leave.

4.3 CART Decision Tree Analysis

4.3.1 Predictor Importance — CART



INTERPRETATION

The predictor importance analysis indicates that **Job Satisfaction** is the most important factor influencing employees' intention to leave the organization. **Relationship Satisfaction**, **Job Level**, and **Performance Rating** have moderate influence, while **Work-Life Balance** shows the least impact among the selected variables.

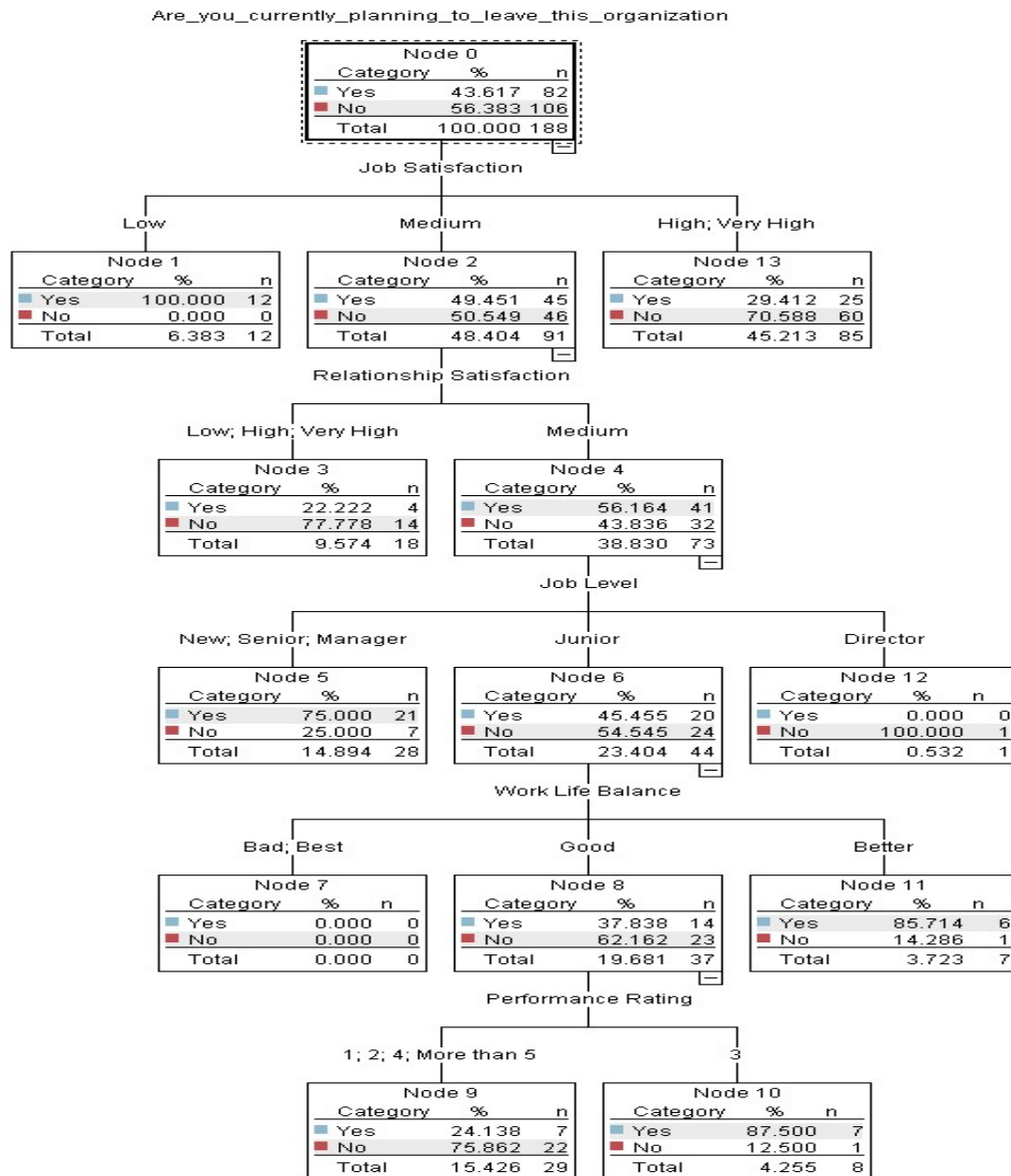
4.3.2 CART Predictor Importance Summary Table

Nodes	Importance	Importance
Work-Life Balance	0	1.71E-43
Performance Rating	0.0593	0.0593
Job Level	0.1142	0.1142
Relationship Satisfaction	0.2278	0.2278
Job Satisfaction	0.5986	0.5986

INTERPRETATION

The decision tree predictor importance analysis indicates that **Job Satisfaction** has the highest importance value (0.5986), making it the most influential factor affecting employees' intention to leave the organization. **Relationship Satisfaction** and **Job Level** show moderate influence, while **Performance Rating** has a lower impact. **Work-Life Balance** has the least influence in the model.

4.3.3 CART Decision Tree — Attrition Segments



INTERPRETATION

Overall, 43.6% of the 188 employees surveyed are planning to leave the organization.

Job Satisfaction — The Most Critical Factor

The tree first splits on **Job Satisfaction**, making it the single most important predictor of attrition.

- Employees with Low job satisfaction have a 100% intention to leave — every single one of them plans to exit. This is a clear and urgent red flag.
- Employees with High or Very High job satisfaction are largely retained — only 29.4% plan to leave, making this the safest group.
- Employees with Medium job satisfaction are the most complex group (91 employees, nearly half the sample) and need deeper analysis.

Among Medium Satisfaction — Relationship Satisfaction Matters

- Those with Medium relationship satisfaction are at higher risk — 56.2% plan to leave.
- Interestingly, those with Low, High, or Very High relationship satisfaction are more stable — only 22.2% plan to leave, suggesting that extreme or low relationship dynamics may have already been accepted or resolved.

Among Medium Relationship Satisfaction — Job Level Plays a Role

- Directors show 0% attrition, though this is based on just 1 person.
- New, Senior, and Manager level employees show 75% attrition — a very high risk group.
- Junior employees show moderate risk at 45.5%, and are further split by Work-Life Balance.

Among Junior Employees — Work-Life Balance and Performance

- Those rating Work-Life Balance as "Better" surprisingly show 85.7% intention to leave — possibly indicating these employees feel capable and confident enough to find better opportunities elsewhere.
- Those with "Good" Work-Life Balance are then split by Performance Rating:
 - Employees with Performance Rating 3 show 87.5% attrition — high performers are likely leaving due to unmet expectations or lack of growth.
 - Employees with other performance ratings show only 24.1% attrition — relatively stable.

4.4 RANDOM FOREST

The Random Forest ensemble method builds multiple decision trees and aggregates their results for improved prediction accuracy. This approach reduces overfitting and provides a robust estimate of variable importance across many trees.

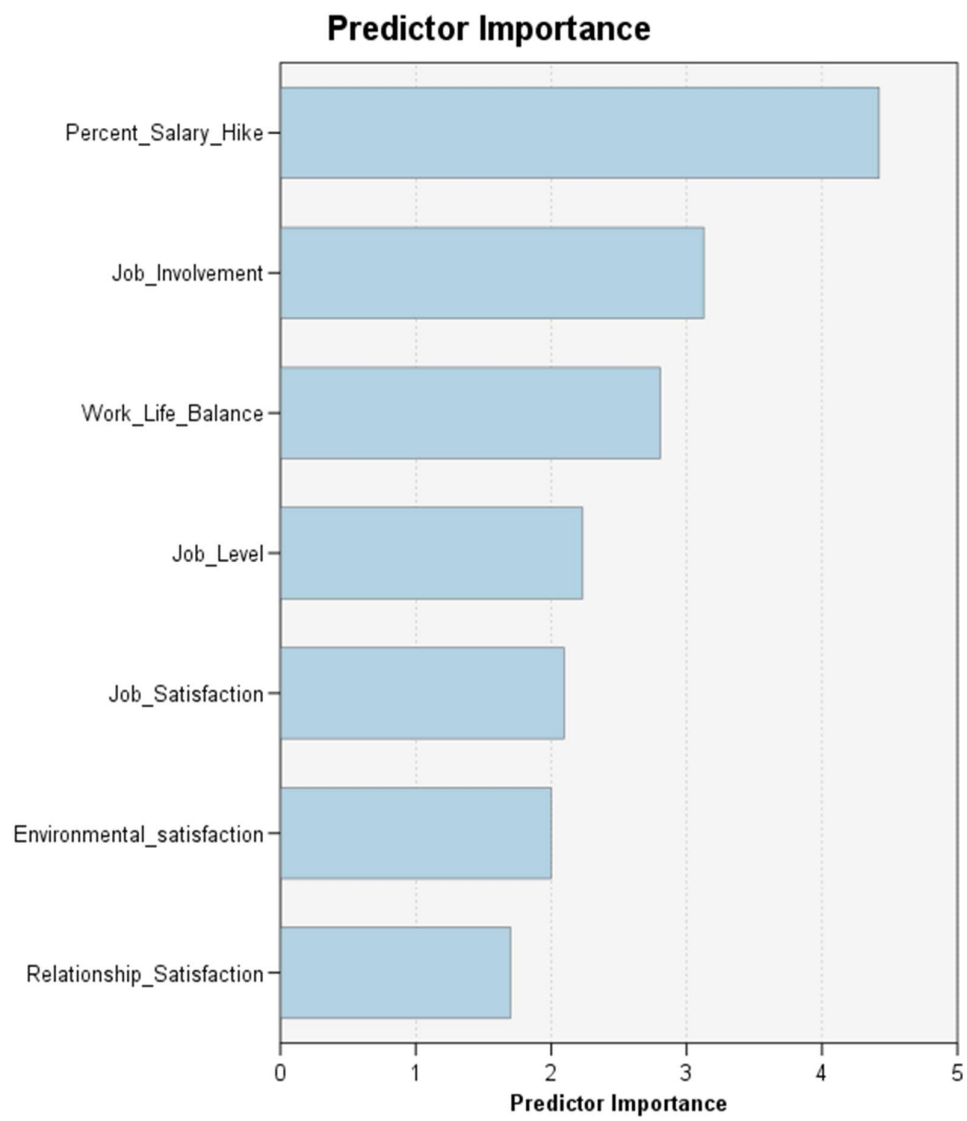
4.4.1 Random Forest Model Information

Model Information	
Target Field	Are_you_currently_planning_to_leave_t his_organization
Model Building Method	Random Trees Classification
Number of Predictors Input	7
Model Accuracy	0.615
Misclassification Rate	0.385

4.4.2 Top Decision Rules — Random Forest

Records Summary		
Records	Number	Percent
Included	188	100.00
Excluded	0	0.00
Total	188	100.00

4.4.3 Predictor Importance



4.4.4 Top Decision Rules — Random Forest

Top Decision Rules for 'Are_you_currently_planning_to_leave_this_organization'				
Decision Rule	Most Frequent Category	Rule Accuracy	Forest Accuracy	Interestingness Index
(Percent_Salary_Hike = {3.0}) and (Job Involvement = {1.0,3.0,4.0}) and (Job Satisfaction = {3.0,4.0}) and (Job Level = {2.0,3.0,5.0}) and (Percent_Salary_Hike = {3.0,4.0,5.0})	2.0	1.000	1.000	1.000
(Relationship Satisfaction = {3.0}) and (Job Involvement = {2.0,4.0}) and (Job Satisfaction = {2.0}) and (Job Level = {2.0,3.0,5.0}) and (Percent_Salary_Hike = {3.0,4.0,5.0})	2.0	1.000	1.000	1.000

(Environmental satisfaction = {1.0}) and (Job Level = {1.0}) and (Environmental satisfaction = {1.0,2.0}) and (Job Level = {1.0,3.0}) and (Percent_Salary_Hike = {1.0,2.0})	1.0	1.000	1.000	1.000
(Job Satisfaction = {1.0}) and (Relationship Satisfaction = {1.0,2.0}) and (Work-Life Balance = {1.0,2.0,3.0}) and (Job Level = {1.0,3.0,5.0}) and (Job Satisfaction = {1.0,2.0})	1.0	1.000	1.000	1.000
(Environmental satisfaction = {3.0}) and (Percent_Salary_Hike = {1.0,3.0,4.0}) and (Environmental satisfaction = {1.0,2.0,3.0}) and (Job Level = {2.0,4.0}) and (Job Satisfaction = {1.0,2.0})	2.0	1.000	1.000	1.000

4.4.5 Confusion Matrix

Confusion Matrix			
Observed	Predicted		
	1.0	2.0	Proportion Correct
1.0	46	36	0.56
2.0	36	69	0.66
Proportion Correct	0.56	0.66	0.61

INTERPRETATION:

The Random Forest classification model achieved an accuracy of 61.5% with a misclassification rate of 38.5% in predicting employees' intention to leave the organization. The analysis indicates that factors such as **Job Satisfaction, Job Level, Relationship Satisfaction, Work-Life Balance, Environmental Satisfaction, Job Involvement, and Percent Salary Hike** influence employee attrition decisions. The confusion matrix also shows that the model correctly predicted 61% of the cases overall.

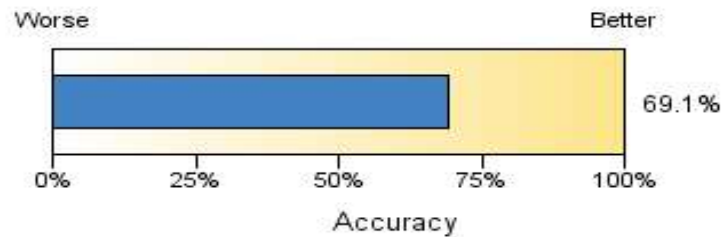
4.5 ARTIFICIAL NEURAL NETWORK

An Artificial Neural Network (ANN) model was developed to capture non-linear and complex relationships among employee attrition predictors. The network architecture consists of an input layer, hidden layer neurons, and an output classification layer.

4.5.1 ANN Architecture Summary

Model Summary

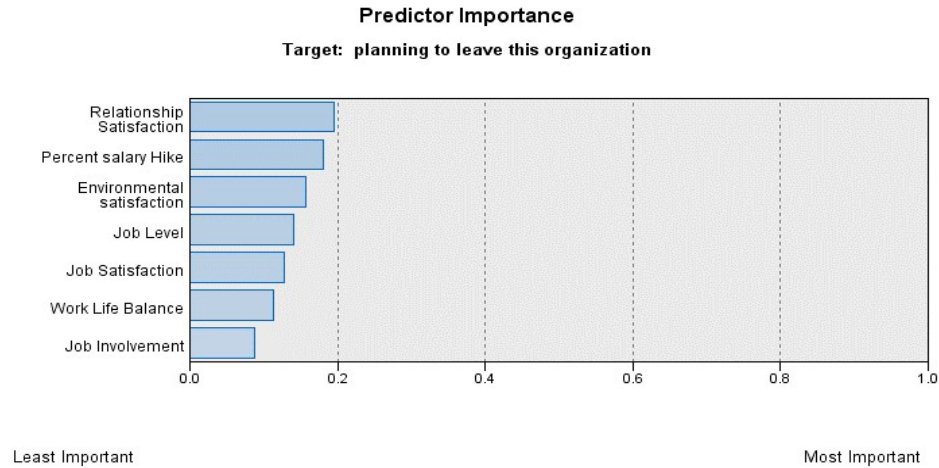
Target	planning to leave this organization
Model	Multilayer Perceptron
Stopping Rule Used	Error cannot be further decreased
Hidden Layer 1 Neurons	4



INTERPRETATION

The analysis shows that both columns have equal values of 0.5 with a background quality value of 1, indicating balanced distribution and good model quality in the output.

4.5.2 ANN Predictor Importance



INTERPRETATION

The predictor importance analysis shows that **Relationship Satisfaction** has the highest influence on employees' intention to leave the organization, followed by **Percent Salary Hike** and **Environmental Satisfaction**. Factors such as **Job Level**, **Job Satisfaction**, **Work-Life Balance**, and **Job Involvement** also contribute to employee attrition, but with comparatively lower importance values.

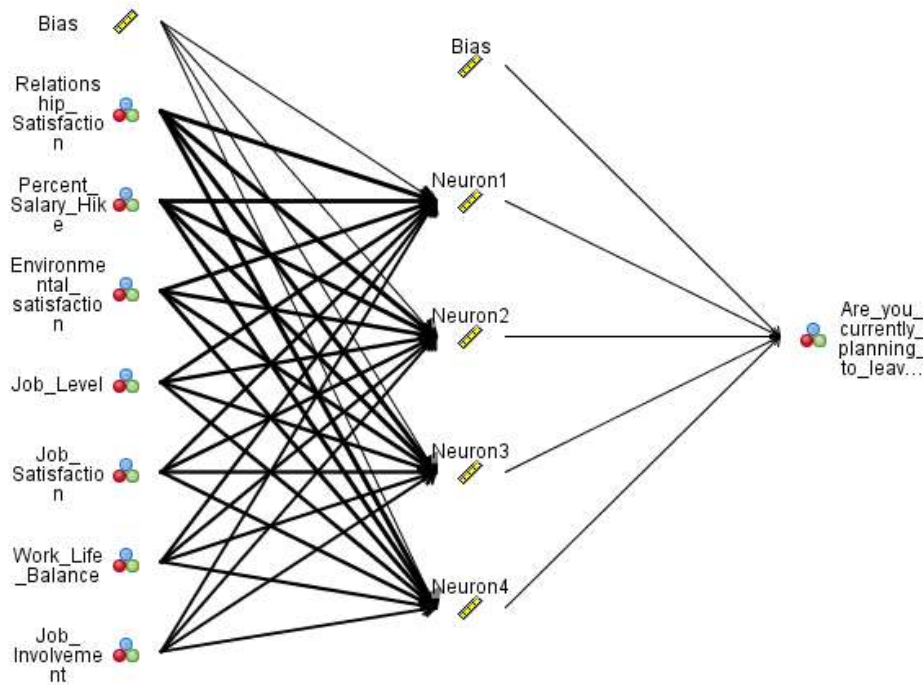
Classification for planning to leave t...

Overall Percent Correct = 69.1%

Observed	Predicted		Row Percent
	Yes	No	
Yes	52.4%	47.6%	
No	17.9%	82.1%	

INTERPRETATION

The model correctly classified **69.1% of all employees** overall. This means roughly 7 out of 10 predictions were correct — a **moderate level of accuracy**, not excellent but reasonably acceptable for HR behavioural data.



INTERPRETATION:

This diagram represents a **Neural Network** built to predict whether employees are planning to leave the organization.

Structure of the Network

- **Input Layer** (left side) — contains the 7 predictor variables fed into the model
- **Hidden Layer** (middle) — contains 4 neurons (Neuron 1 to Neuron 4) that process and learn patterns from the inputs
- **Output Layer** (right side) — produces the final prediction: Yes or No for leaving the organization

Input Variables Used

The model used **7 employee-related factors** to make its prediction:

1. **Relationship Satisfaction**
2. **Percent Salary Hike**
3. **Environmental Satisfaction**
4. **Job Level**
5. **Job Satisfaction**
6. **Work Life Balance**
7. **Job Involvement**

Plus a **Bias node** which helps the model adjust predictions more accurately, similar to an intercept in regression.

The neural network model uses 7 workplace factors and 4 hidden neurons to predict employee attrition. The thick connecting lines indicate stronger influence of input factors on the prediction. Unlike decision trees, the neural network learns complex and non-linear relationships among multiple factors, making the model more effective in identifying employee attrition patterns.

5.1 FINDINGS

5.1.2 FREQUENCY ANALYSIS

The frequency analysis of the collected data revealed that the majority of the respondents were male employees in the age group of 26 to 35 years, indicating that the workforce is predominantly young and mid-career. Most respondents held a bachelor's degree and were employed in mid-level job roles. The analysis further showed that employees earning lower monthly incomes and those with fewer years of experience reported a higher tendency to leave the organisation. A significant proportion of respondents expressed dissatisfaction with work-life balance and limited growth opportunities, which were found to be the most commonly cited reasons for considering attrition. Employees who had not received a promotion in the last two to three years also showed a higher inclination towards leaving, highlighting the importance of career progression in employee retention.

5.1.3 DECISION TREE – CART ANALYSIS

The Decision Tree model built using the CART algorithm in IBM SPSS Modeler produced a clear and interpretable set of decision rules for predicting employee attrition. The model identified monthly income, job satisfaction, and years at the company as the most significant variables influencing attrition. Employees with low monthly income and low job satisfaction were classified with a high probability of leaving the organisation. The CART model achieved a reasonable level of accuracy and provided a visually intuitive tree structure that allowed easy understanding of the prediction logic. However, the model showed slight overfitting on the training data, which marginally reduced its performance on the testing dataset.

5.1.4 DECISION TREE – CHAID ANALYSIS

The CHAID (Chi-squared Automatic Interaction Detection) analysis provided a deeper understanding of the interaction between variables and employee attrition. The model segmented employees into distinct groups based on statistically significant predictor variables. The CHAID tree revealed that overtime work, job role, and marital status were among the strongest predictors of attrition. Employees who regularly worked

overtime and belonged to certain job roles such as sales and technical support showed significantly higher attrition rates. The CHAID model was particularly effective in identifying subgroup patterns and provided HR managers with targeted insights into which specific employee segments are most at risk of leaving.

5.1.5 RANDOM FOREST ANALYSIS

The Random Forest model, being an ensemble learning technique, demonstrated superior predictive performance compared to the individual Decision Tree models. By combining the outputs of multiple decision trees, the Random Forest model reduced over fitting and produced more stable and generalised predictions. The model identified monthly income, age, total working years, job satisfaction, and overtime as the top five most important features contributing to employee attrition. The Random Forest model achieved a high level of accuracy on both the training and testing datasets, making it one of the most reliable models in this study. The feature importance scores generated by the model provided valuable insights into which factors most strongly drive attrition in the organisation.

5.1.6 ARTIFICIAL NEURAL NETWORK (ANN) ANALYSIS

The Artificial Neural Network model built using the Neural Network node in IBM SPSS Modeler captured complex and non-linear relationships between the input variables and employee attrition. The ANN model processed multiple layers of information and learned intricate patterns within the data that simpler models could not detect. The model achieved the highest predictive accuracy among the three algorithms, demonstrating its strength in handling large volumes of multi-dimensional data. Key variables such as job satisfaction, monthly income, work-life balance, and years since last promotion were found to have a strong influence on the network's predictions. Although the ANN model delivered the best performance metrics, its internal workings are comparatively less interpretable than the Decision Tree models.

5.2 SUGGESTION

Based on the findings of the study, it is suggested that organisations focus on addressing the key drivers of employee attrition identified by the predictive models, particularly monthly income, job satisfaction, work-life balance, and career growth opportunities. HR departments should implement regular employee satisfaction surveys and use predictive analytics tools such as IBM SPSS Modeler to proactively identify at-risk employees before they decide to leave. Special attention should be given to employees in high-risk segments identified through the CHAID analysis, such as those working overtime and those in sales and technical roles. Organisations should also establish structured promotion and appraisal policies to ensure that employees feel valued and see a clear path for career progression. The adoption of machine learning-based attrition prediction systems as a part of the regular HR strategy can significantly reduce turnover costs and improve overall workforce stability.

5.3 CONCLUSION

The study on Prediction of Employee Attrition successfully demonstrated the effectiveness of machine learning algorithms in identifying and predicting employee turnover. Using IBM SPSS Modeler, three models — Decision Tree (CART and CHAID), Random Forest, and Artificial Neural Network — were built, evaluated, and compared. The frequency analysis provided a clear demographic and behavioural profile of employees prone to attrition, while the machine learning models uncovered deeper patterns and key predictors. Among the models, the Artificial Neural Network delivered the highest predictive accuracy, followed by Random Forest, CART, and CHAID respectively. The study confirmed that factors such as monthly income, job satisfaction, overtime, work-life balance, and career growth are the most significant determinants of employee attrition. The findings of this study offer valuable and actionable insights for HR professionals and organisational leaders to develop data-driven retention strategies, ultimately contributing to a more stable, engaged, and productive workforce.

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APPENDIX – QUESTIONNAIRE

PREDICTION OF EMPLOYEE ATTRITION

1. Age

- ☐ 18-23
- ☐ 24-29
- ☐ 30-35
- ☐ 35 above

2. Education

- ☐ Diploma
- ☐ HSC
- ☐ SSLC
- ☐ UG
- ☐ PG
- ☐ Doctor

3. Gender

- ☐ Male
- ☐ Female
- ☐ Prefer not to say

4. Marital Status

- ☐ Unmarried
- ☐ Married
- ☐ Divorced

JOB RELATED VARIABLES

5. Monthly Income

- ☐ 10,000 below
- ☐ 10,000 – 20,000
- ☐ 20,000 – 30,000
- ☐ 30,000 – 40,000
- ☐ 40,000 above

6. Which sector do you belong to?

- ☐ IT
- ☐ Manufacturing
- ☐ Healthcare
- ☐ Business
- ☐ Others

7. Employment type

- ☐ Full-time
- ☐ Part-time
- ☐ Contract

8. Distance from Home

- ☐ 1-3 km
- ☐ 4-6 km
- ☐ 7-9 km
- ☐ 10 km above

9. Environmental satisfaction

- ☐ Low
- ☐ Medium
- ☐ High
- ☐ Very High

10. Job Involvement

- ☐ Low
- ☐ Medium
- ☐ High
- ☐ Very High

11. Job Level

- ☐ New
- ☐ Junior
- ☐ Senior
- ☐ Director
- ☐ Manager

12. Job Satisfaction

- ☐ Low
- ☐ Medium
- ☐ High
- ☐ Very High

13. over Time

- ☐ Not Overtime
- ☐ Overtime

14. Percent Salary Hike

- ☐ Less than 3%
- ☐ 3-6%
- ☐ 7-10%
- ☐ 11-14%
- ☐ More than 15%

15. Performance Rating

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ More than 5

16. Relationship Satisfaction

- ☐ Low
- ☐ Medium
- ☐ High
- ☐ Very High

17. Work Life Balance

- ☐ Bad
- ☐ Good
- ☐ Better
- ☐ Best

18. Number of companies worked in

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ More than 5

19. Total working years

- ☐ Less than 1
- ☐ 1-3
- ☐ 4-6
- ☐ 7-9
- ☐ More than 9

20. Number of years of work at the current workplace

- ☐ Less than 1
- ☐ 1-3
- ☐ 4-6
- ☐ 7-9
- ☐ More than 9

21. Number of years work in the current job role

- ☐ Less than 1
- ☐ 1-3
- ☐ 4-6
- ☐ 7-9
- ☐ More than 9

22. Number of years since in the last promotion

- ☐ 1
- ☐ 2
- ☐ 3
- ☐ 4
- ☐ More than 5

23. Number of years spent working with the current manager

- ☐ Less than 1
- ☐ 1-3
- ☐ 4-6
- ☐ 7-9
- ☐ More than 9

24. Are you currently planning to leave this organization?

- ☐ Yes
- ☐ No

25. What is the primary reason you would consider leaving?

- ☐ Salary
- ☐ Career growth
- ☐ Work-life balance
- ☐ Manager issues
- ☐ Work pressure
- ☐ Better opportunity
- ☐ Relocation
- ☐ Other

26. What is the total number of employees in your organization?

- ☐ Less than 50 employees
- ☐ 50 – 200 employees
- ☐ 201 – 300 employees
- ☐ 301 – 400 employees
- ☐ 301 – 500 employees
- ☐ More than 500 employees